Lab 5 - Part 1

Welcome to the Wireless World

Course: RoboLink: Build and Drive Your Smartphone-Controlled Robot
Program: Mini Course Program 2026, Carleton University
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Lab Mission
In this lab, you will test wireless communication between an Android phone and an Arduino Nano. The phone app sends a command over Bluetooth Low Energy (BLE). The BLE module passes the command to the Arduino, and the Arduino shows that it received the command by printing a message and turning the built-in LED on or off.

Group Member Names ______________________________

Date ______________________________

1. Learning Goals

By the end of this lab, you should be able to:

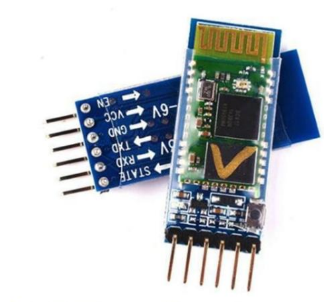
- Connect a Bluetooth/BLE module to an Arduino Nano ATmega168.
- Explain the difference between USB Serial communication and wireless communication.
- Install and use an Android app to send wireless commands.
- Use the Arduino Serial Monitor to check what command the Arduino received.
- Explain why the module RXD pin should be protected with a voltage divider.
- Troubleshoot wireless connection problems using careful observations.

2. Big Idea

Wireless control is still serial communication.
In Lab 4, commands came from the computer Serial Monitor through the USB cable. In this lab, commands come from a phone app through a Bluetooth/BLE module. The Arduino still receives simple command values such as `1`, `2`, `3`, `4`, and `5`.

3. Important Note About This Module

Special BLE Edition
The module used in this lab looks like an HC-05 Bluetooth module, but it is a **special edition that works with BLE, not SPP**. SPP means Serial Port Profile, which is used by many classic Bluetooth HC-05 modules. This lab uses BLE, so students must use the provided Android app instead of a normal classic Bluetooth terminal app.



Bluetooth/BLE module used in this lab. The pins are labelled EN, VCC, GND, TXD, RXD, and STATE.

4. Materials

Arduino Nano ATmega168	Receives wireless commands and runs the test code.
Special BLE HC-05-style module	Receives commands from the Android phone app.
Android phone	Runs the provided mobile app. iPhone is not supported in this lab.
USB cable	Uploads code and opens the Serial Monitor.
Jumper wires	Connect the module to the Arduino Nano.
Voltage divider resistors	Protect the module RXD pin from the Arduino's 5V TX signal.

5. Install the Android App

Android Only
The mobile app for this lab works on **Android phones only**. It is not designed for iPhone.

Before Scanning for the Robot
Make sure **Bluetooth** and **Location/GPS** are both turned on. The app must also have the required Android permissions for connectivity and location, such as Nearby Devices, Bluetooth, and Location permissions. Without these permissions, the app may not find the BLE module even if the wiring is correct.

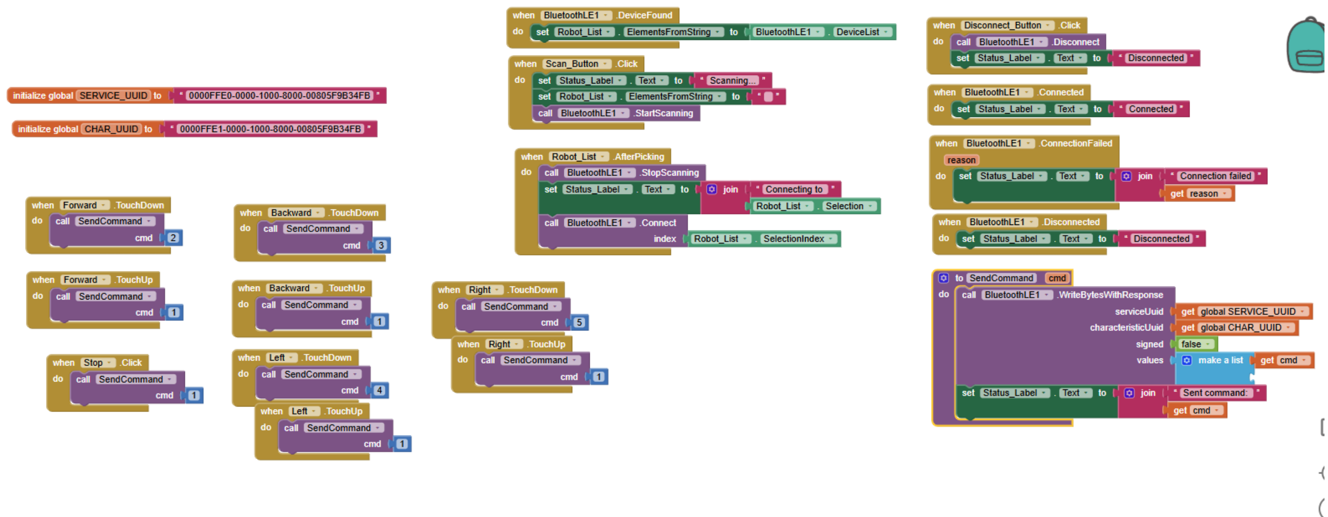
Install the app from this link:

https://drive.google.com/file/d/1OtnT_rNqe7tGeGTrUGqdO2R689uXVynw/view?usp=drive_link

1. Open the link on an Android phone.
2. Download the app file.
3. If Android asks for permission to install from this source, ask your instructor before continuing.
4. Install the app.
5. Open the app and keep it ready for the testing steps.

6. How the App Sends Commands

The app was created using MIT App Inventor. The app buttons send small command values to the BLE module.



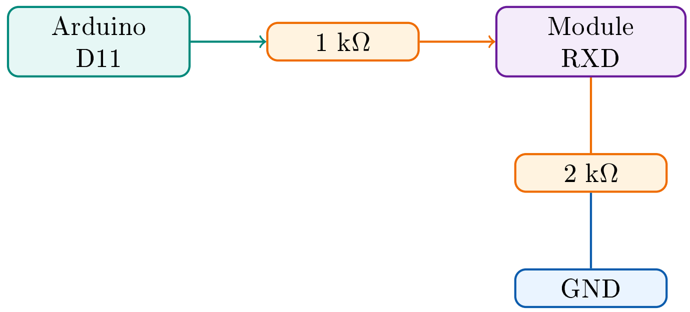
MIT App Inventor blocks. The app scans for BLE devices, connects to the robot module, and sends command values when buttons are pressed.

Command	Meaning	In This Test
1	Stop	LED OFF
2	Forward	LED ON
3	Backward	LED ON
4	Left	LED ON
5	Right	LED ON

7. Wiring the BLE Module

Protect the RXD Pin
The Arduino Nano sends 5V logic from D11. Many Bluetooth/BLE module RXD pins expect about 3.3V logic. Use a voltage divider between Arduino D11 and module RXD.

Module Pin	Arduino Nano Pin	Notes
TXD	D10	Module transmits to Arduino receive pin.
RXD	D11	Use a voltage divider before RXD.
GND	GND	Must share ground with the Arduino.
VCC	5V or 3.3V	Use the correct power for your module board.
EN	Not used	Leave unconnected for this lab.
STATE	Not used	Leave unconnected for this lab.



One simple voltage divider: Arduino D11 → 1 kΩ → module RXD, then module RXD → 2 kΩ → GND.

8. Arduino IDE Setup

1. Open the Arduino IDE or PlatformIO project.
2. Open the Lab 5 Part 1 code.
3. Select **Tools** → **Board** → **Arduino AVR Boards** → **Arduino Nano**.
4. Select **Tools** → **Processor** → **ATmega168**.
5. Select the correct **Tools** → **Port**.
6. Upload the code.
7. Open **Tools** → **Serial Monitor**.
8. Set the baud rate to **9600**.

9. Experiment Procedure

Part A: USB Test First

1. Open the Serial Monitor at 9600 baud.
2. Send `?`.
3. Send `2`.
4. Send `1`.

Observation: What did the Serial Monitor print? What happened to the LED?

Part B: Connect the Phone App

1. Turn on Bluetooth and Location/GPS on the Android phone.
2. Check that the app has Bluetooth/connectivity and Location permissions.
3. Open the RoboLink app.
4. Press the scan button in the app.
5. Choose the BLE module from the list.
6. Wait until the app says it is connected.

Connection observation: What name appeared for your BLE module?

Part C: Wireless Command Test

Press each app button and record what happens.

Command	Expected Result	What Actually Happened?
1 / Stop	LED turns OFF	
2 / Forward	LED turns ON	
3 / Backward	LED turns ON	
4 / Left	LED turns ON	
5 / Right	LED turns ON	

10. What Is Happening in the Code?

`SoftwareSerial`	Creates a second serial port on pins D10 and D11.
`bluetooth.begin(9600)`	Starts communication with the BLE module.
`bluetooth.available()`	Checks if the phone app sent a command.
`bluetooth.read()`	Reads one command from the BLE module.
`normalizeCommand()`	Makes raw byte commands, text digits, and lowercase letters easier to handle.
`handleCommand()`	Chooses what action to run for each command.
`digitalWrite(13, HIGH)`	Turns the built-in LED on.
`digitalWrite(13, LOW)`	Turns the built-in LED off.

11. Expected Problems and Fixes

Problem	Possible Cause	What To Try
Phone cannot install app	Phone is not Android, or install permission is blocked	Use an Android phone and ask the instructor about install permission.
App cannot find module	BLE module is not powered, Bluetooth is off, Location/GPS is off, or app permissions are missing	Check VCC, GND, Bluetooth, Location/GPS, and app permissions.
App connects then fails	Wrong module type or weak power	Confirm this is the BLE special edition module and check power.
Serial Monitor shows nothing	Wrong baud rate or wrong port	Set Serial Monitor to 9600 and check Tools → Port.
Upload fails	Wrong Nano processor setting	Select Tools → Processor → ATmega168.
Commands do nothing	TXD/RXD wiring is swapped or loose	Module TXD goes to D10. Arduino D11 goes to module RXD through voltage divider.
Module gets hot	Wiring or voltage is wrong	Disconnect power and ask the instructor to check the circuit.

12. Reflection Questions

Answer in full sentences.

1. Why do we test wireless communication with the LED before connecting the motors?
2. What is the difference between sending a command by USB Serial Monitor and sending it by BLE?
3. Why does the Arduino D11 signal need a voltage divider before going to the module RXD pin?
4. Why does this lab need the provided Android BLE app instead of a normal classic Bluetooth terminal app?
5. If the app says it is connected but the Arduino receives nothing, what are two things you would check?

13. Exit Ticket

In one or two sentences: What did you learn about how a phone can send commands to a robot?

14. Teacher Check

Teacher Checklist

- [ ] **Wiring checked** Notes: ____________________
- [ ] **Code uploaded** Notes: ____________________
- [ ] **Android app installed** Notes: ____________________
- [ ] **BLE module connected in app** Notes: ____________________
- [ ] **Arduino receives wireless commands** Notes: ____________________
- [ ] **Students completed reflection questions** Notes: ____________________